Measurement Science

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Measurement Science (A06761)

Short Title:	Measurement Science		
Faculty	Science and Computing		
Department	Science		
Cluster	Mathematics and Physics		
Author	GGALLAGHER		
Credits	10	**Level:**	Introductory

Description of Module / Aims

This is an introductory module in measurement science. Much of the content is basic physics, providing the student with a background to understand the more applied measurement concepts considered towards the end of the module. No previous knowledge of physics or measurement science is assumed.

Programmes

PHYS-0018	Higher Certificate in Science in Good Manufacturing Practice and Technology (WD_SGMPT_C)	1 / 2 / M

Indicative Content

- Introduction: SI system of units; scientific notation; simple calculations using physical quantities
- Basics mechanics: vectors and scalars; Newtons laws of motion and gravitation
- Heat and temperature: specific heat capacity and latent heat
- Waves motion and light: electromagnetic waves and electromagnetic spectrum
- Introduction to electricity: simple circuits; resistors in series and parallel; electrical safety
- Analogue and digital signals: temperature sensors
- Measurement science: systematic and random errors; strain gauge and other transducers; calibration, standards, traceability ladder

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Use the SI system of units and perform standard types of formula-based calculations in basic physics.
2. Define basic terms in physics and describe applications of the basic physics presented.
3. Explain the principles behind a wide range of temperature sensors.
4. Calculate important quantities associated with simple electrical circuits.
5. Use laboratory equipment in a safe and accurate manner.
6. Describe the basic components of a measurement system and outline important measurement principles.
7. Report on and critically assess experiments carried out to demonstrate principles of physics in a laboratory context.

Learning and Teaching Methods

- Lectures.
- Practical programme.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3,4,6
Continuous Assessment	50%	
Practical	50%	5,7

Assessment Criteria

<40%: The student has failed to demonstrate a reasonable grasp of the material and is unable to carry out routine calculations reliably. Experimental work is generally poor.

40%-49%: Able to demonstrate a basic understanding of the fundamental concepts outlined in the indicative content. Able to provide partial solutions to numerical problems. Practical work is usually of an acceptable quality.

50%-59%: Able to clearly describe and define concepts encountered in the indicative content. Able to analyse physical situations and apply suitable problem-solving techniques to find numerical solutions. Carries out practical work capably and writes up same to an acceptable standard.

60%-69%: In addition to the above, able to apply problem-solving techniques to new similar problems. Demonstrates a good understanding of the majority of the course content. Practical tasks are performed and written-up well.

70%-100%: All the previous to an excellent level. Demonstrates an ability to put a numerical solution into a context and assess whether such solutions are meaningful. Experimental work is carried out to an high standard. Reports are clear and insightful with the student demonstrating an ability to reflect on the strengths and weaknesses of the experimental techniques used.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture		24
Independent Learning		246

Supplementary Material

- Johnson, K. _Physics for You_. UK: Stanley-Thornes, 2011.
- O'Regan, D. _Real-World Physics_. Dublin: Folens, 2000.
- Squires, G.L. _Practical Physics_. UK: Cambridge University Press, 2008.

Requested Resources

- SCIENCE LAB: Physics